# 3.3 SAND

## Learning Objectives

**At the end of this lesson you shall be able to**

- Define river sand and manufactured sand (M-sand).
- Compare M-sand and river sand.

TDD584

### 3.3.1 Introduction

Sand is a building material used in construction for preparing mortar, concrete and also for filling under floor, basements. It is technically named as fine aggregate. It is used in concrete to fill up the voids leaved by coarse aggregates.

Now-a-days, it is practically impossible to get river sand in large quantities. Hence, M sand (manufactured sand) is introduced in this field to overcome this deficiency.

### 3.3.2 River Sand

Sand is generally composed of rounded particles and may or may not contain clay or other impurities. It is obtained from the banks and beds of rivers.

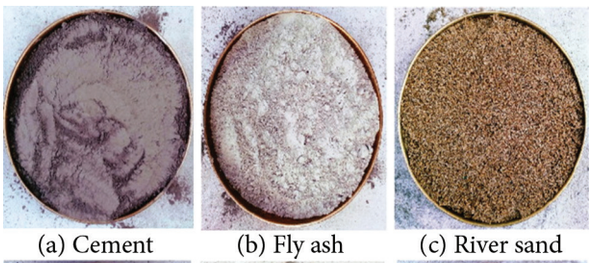


### 3.3.3 Manufactured Sand (M Sand)

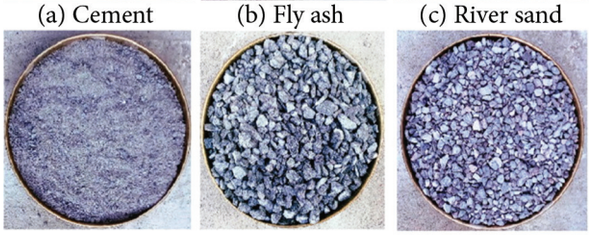
M Sand is defined as a crushed fine aggregate produced from broken granite blocks. Production of M sand generally involves crushing, screening and possible washing.



**Search link:** http://www.youtube.com/watch/m-sand



(a) Cement (b) Fly ash (c) River sand



(d) M-Sand (e) 19mm Coarse aggregate (f) 6mm - 12mm aggregate



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